



Visual Recognition using Deep Learning

2026 Spring, Homework 4

Release Date: 2026/05/12 15:40

Homework 4

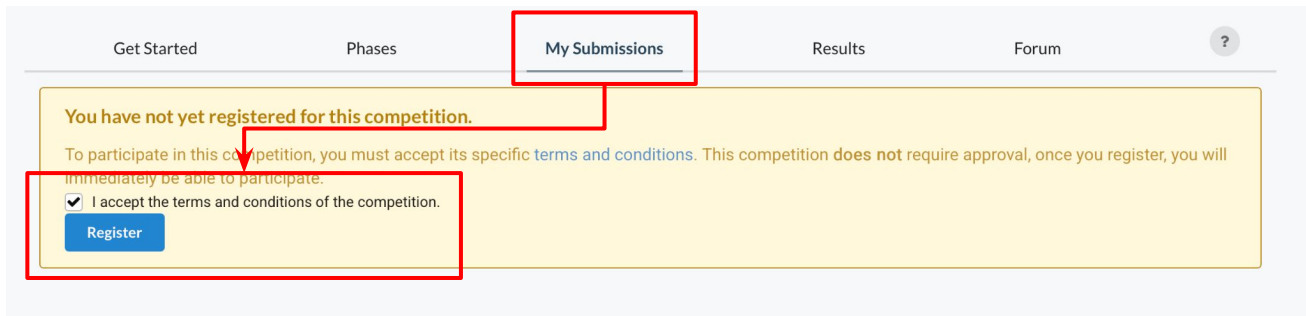
- **Deadline:** 23:59, 06/02, 2026
- **Participate the competition (85%):** Image Restoration
 - Participant the competition on the CodaBench and get the highest score as possible. (80%)
 - Code reliability & quality: GitHub (5%)
- **Report and code (15%):** Document your method and findings.
 - **Report**
 - **In PDF format and written in English.** (5pt penalty)
 - Introduction to your method (e.g., data pre-processing, model architecture, hyper-parameters)
 - Innovative ideas or additional experiments to further improve the model.
 - **Code**
 - Zip your code (.py) alone with report - Submit to E3.
 - You should also put your code on your GitHub repository and provide the link in the report.

Links

- [Link to the dataset, sample code and example submission](#)
- [Link to the competition](#)

How to participate the competition and do submission

1. Register an account on [CodaBench](#). (**Do not** use two accounts to participate in the competition.)
2. After you click the competition link, go to My Submissions, and join the competition.



The screenshot shows the 'My Submissions' tab selected in a navigation bar. Below the navigation bar, a yellow box contains the following text: 'You have not yet registered for this competition.' Below this, it says: 'To participate in this competition, you must accept its specific terms and conditions. This competition does not require approval, once you register, you will immediately be able to participate.' At the bottom of the yellow box, there is a checked checkbox labeled 'I accept the terms and conditions of the competition.' and a blue 'Register' button. Red boxes and arrows highlight the 'My Submissions' tab, the registration message, and the checkbox/button area.

Get Started Phases **My Submissions** Results Forum ?

You have not yet registered for this competition.

To participate in this competition, you must accept its specific terms and conditions. This competition does not require approval, once you register, you will immediately be able to participate.

☒ I accept the terms and conditions of the competition.

Register

How to participate the competition and do submission

3. Enter your student ID number, then upload the submission file.

The screenshot shows a web interface for a competition. At the top, there are navigation tabs: 'Get Started', 'Phases', 'My Submissions' (highlighted with a red box), 'Results', and 'Forum'. Below the tabs, there's a 'Testing Phase' button. Two statistics are displayed: 'Number of submissions used for the day' (2 out of 5) and 'Number of total submissions used' (2 out of 99). The main section is titled 'Submission upload'. Underneath, there's a 'Metadata or Fact Sheet' section. A red box highlights the 'STUDENT ID: *' input field, with a red arrow pointing to it from the 'My Submissions' tab. Below this, there's a 'Submit as:' dropdown menu with 'yourself' selected. A red box highlights the file upload icon (a document with a plus sign) in the bottom left corner, with a red arrow pointing to it from the 'STUDENT ID' field.

No limitation for the name of the “zip” file;
however, inside the zip file, your **result file must
be named “pred.npz”**

How to participate the competition and do submission

4. Don't forget to “Add to Leaderboard”.

5. Don't forget to check your results on the leaderboard.

ID # ▾	File name	Date	Status	Score	
569695	prediction.csv.zip	2026-02-27 21:17	Finished	0.01	Add to Leaderboard 📁 🔗 🗑️
569689	prediction.csv.zip	2026-02-27 21:16	Finished	0.01	✅ 🔗

Get Started	Phases	My Submissions	Results	Forum	?
Testing Phase					
Filter Leaderboard by Columns					
Results					
Task:				Fact Sheet Answers	Main classification Task
#	Participant	Date	ID	STUDENT ID	Prediction score (Public)
🏆	tltsai1	2026-02-27 21:16	569689	123456789	0.01

Coding Environment

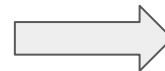
- Recommendation: Python 3.9 or higher
- Tips
 - We recommend you to use **virtual environments** when implementing your homework assignments.
 - Here are some popular virtual environment management tools
 - [Poetry](#)
 - [Conda](#)
 - [Virtualenv](#)

Numpy & PyTorch

- Numpy Tutorial: [Link](#)
- PyTorch Tutorial: [Link](#)
 - Free to use any modules and functions

Task and Dataset

- Task type: Image Restoration
- Dataset
 - Degraded images (2 types: Rain and Snow)
 - Training / Validation:
 - 1600 degraded images per type
 - 1600 corresponding clean images per type
 - Test:
 - 50 degraded images per type
- Target
 - Clean images corresponding to each degraded image
- Evaluation
 - PSNR (Peak Signal-to-Noise Ratio)



Task Requirement and Limitations

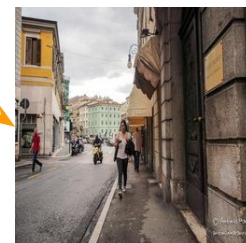
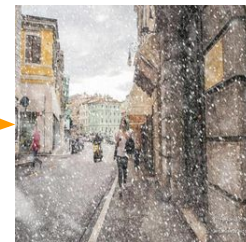
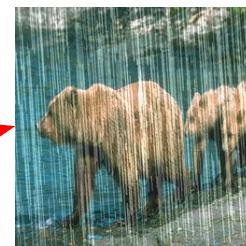
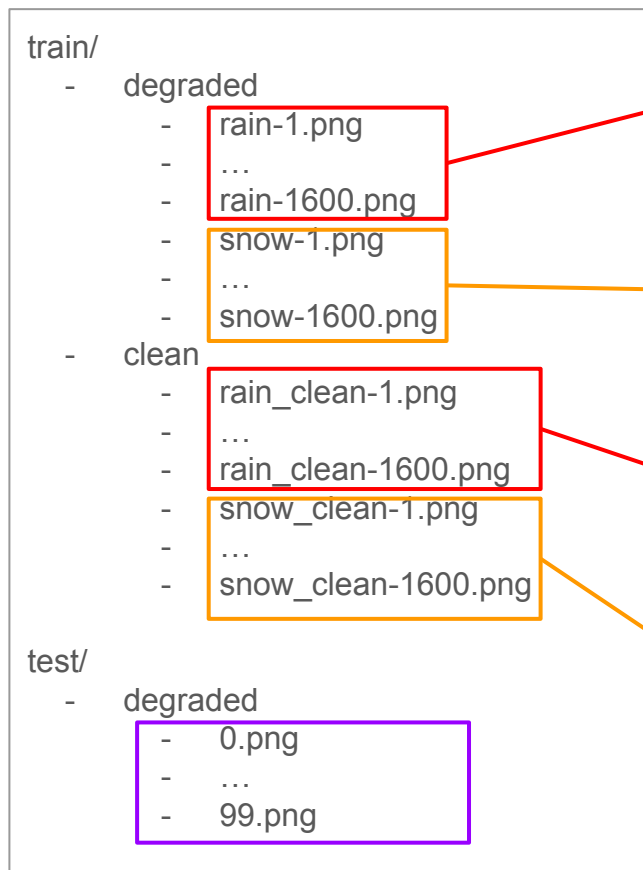
- **Requirement:** Train a single model capable of restoring both types of degraded images (Rain and Snow).
- **Limitations**
 - **No** external data (i.e., data from other sources) is allowed.
 - Pretrained weights are **NOT** allowed — the model must be trained from scratch.
 - You are required to use PromptIR as the model in this task.
 - You may modify components/modules of PromptIR to improve performance. In your report, you must:
 - Elaborate on the key design or contribution of PromptIR.
 - Explain the modifications you made to improve the model performance.
 - Cite the paper properly.

Dataset Inspection

The training set contains degraded images and clean images.

The degraded images include rain images and snow images, such as **rain-1.png** to **rain-1600.png** and **snow-1.png** to **snow-1600.png**.

Each degraded image has a matching clean image, such as **rain_clean-1.png** to **rain_clean-1600.png** and **snow_clean-1.png** to **snow_clean-1600.png**.



The test set is different. Its images use simple number names, such as 0.png to 99.png. These names do not show whether the image is degraded by rain or snow.

Output Format Example for Image Restoration

You are required to submit a file named `pred.npz`. This file should store all restored images in a dictionary-like format using NumPy.

- Keys are the filenames of the original images (e.g., '0.png', '1.png', ...)
- The values are the restored images. Each image should be stored as a NumPy array with shape (3, H, W).
 - Shape explanation:
 - 3 channels (Red, Green, Blue)
 - Height and Width match the original image size
 - Values can be uint8 (0–255)
- Submission may take some time, so it is better to prepare early and submit on time.

```
import os
import numpy as np
from PIL import Image

# Set your image folder path
folder_path = '/path/to/your/folder'
output_npz = 'pred.npz'

# Initialize dictionary to hold image arrays
images_dict = {}

# Loop through all files in the folder
for filename in os.listdir(folder_path):
    if filename.lower().endswith(('.png', '.jpg', '.jpeg')):
        file_path = os.path.join(folder_path, filename)

        # Load image and convert to RGB
        image = Image.open(file_path).convert('RGB')
        img_array = np.array(image)

        # Rearrange to (3, H, W)
        img_array = np.transpose(img_array, (2, 0, 1))

        # Add to dictionary
        images_dict[filename] = img_array

# Save to .npz file
np.savez(output_npz, **images_dict)

print(f"Saved {len(images_dict)} images to {output_npz}")
```

Grading Policy - Report (15%)

- Format: PDF, written in English. (-5pts if not followed)
- Sections that you should include
 - **Introduction** to the task and core idea of your method
 - **Method**: Describe your data preprocessing procedure, model architecture, hyperparameter settings, and any modifications or adjustments you made.
 - You need to describe each key component in your model. (e.g., Backbone: ResNet, Neck: FPN, Heads: Mask/Transformer decode, etc.)
 - **Results**: your findings / model performance / visualization (e.g., training curve, confusion matrix, etc.)
 - **References**: Your method references (paper / Github sources, must include if you use any). You may refer to the [ECCV 2026 template](#) for the citation and reference formatting guidelines.

12pts

We encourage you to stand on the shoulders of giants - only clone and run it is not enough.

3pts

- **Additional experiments** to explore better performance
 - Simply tuning the hyper-parameters doesn't count (e.g., batch-size, LR, different optimizers).
 - Hint: Try to add/remove some layers, use different loss functions, etc.
- You should 1) include your hypothesis (why you do this), 2) How this may (or may not) work, and 3) The experiment results and their implications.

Grading Policy - Code Reliability (5%)

1. Please follow the PEP8 instructions and lint your code.
2. Push your code to the GitHub
 - It should contain a README.md file that includes the following sections:
 - Introduction
 - Environment Setup
 - Usage
 - Performance Snapshot

Python Coding Style Guide Reference

1. [PEP8](#)
2. [Google Python Style](#)

NYCU Computer Vision 2026 HW1

- Student ID: 123456789
- Name: Your Name

Introduction

Briefly introduce your work here.

Environment Setup

How to install dependencies.

```
pip install -r requirements.txt
```

Usage

Training

How to train your model.

```
python train.py
```

Inference

How to run inference.

```
python inference.py
```

Performance Snapshot

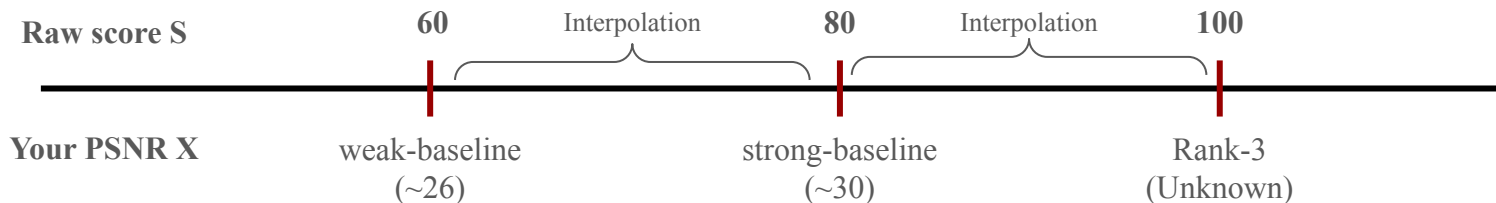
Insert a screenshot of the leaderboard here.

Grading Policy - Competition (80%)

We will use private (hidden) leaderboard to evaluate the performance (the distribution is similar for data in public and private set.) The public leaderboard is for you as reference.

Your score (competition):

- **Less than weak-baseline:** $S = 0$
- **Between weak-baseline and strong baseline:** $60 + (X - \text{w.baseline}) / (\text{s.baseline} - \text{w.baseline}) * (80 - 60)$
- **Between strong-baseline and Rank3:** $80 + (X - \text{s.baseline}) / (\text{rank3} - \text{s.baseline}) * (100 - 80)$
- **Rank1, 2, 3:** 100



Submission

- Compress your code and report into a **.zip file** and submit it to E3.
 - Don't forget to push your code to GitHub. And **your GitHub link should be written in the report.**
- Report should be written **in English.**
- <STUDENT ID>_HW4.zip
 - codes (.py, folders, etc)
 - <STUDENT ID>_HW4.**pdf** (NO .doc, .docx or others format)
- **Don't put the data (e.g. x.jpg / train.csv / test.csv) and model checkpoints into submission file. (-5pts if not followed)**

Other rules

- **Late Policy**: A penalty of **20 points** per additional late day. (-20pts per delayed day)
 - For example, If you get 90 points but delay for two days, your will get only 50 points!
- **No Plagiarism**: You should complete the assignment by yourself. Students engaged in plagiarism will be penalized heavily. Super serious penalty.
 - e.g. -100pt for the assignment or failed this course, etc
 - Report to academic integrity office

FAQs

- Can I use any library/package/framework from GitHub or other resources?
 - Yes, we encourage you to learn how to leverage existing knowledge on your own task
 - e.g., Github of published works and model zoo from Torchvision
 - Focus on how to step forward from them - That's why part of scores comes from your competition ranks
 - You should not copy-and-paste from your classmates (Plagiarism)
- How to handle the GPU Out-of-Memory (OOM) issue?
 - Easy answer - Make your batch size smaller or make your model smaller.
 - Advanced methods: Try to figure it out by yourself. (Many online resources and AI-assistance)

FAQs

- If I don't have my own GPU – Use Google Colab
 - It should be 12 hours, please check [this discussion in the stackoverflow](#)
 - And some tricks [here](#) may make it longer.
- If you have other questions, ask on [E3 forum](#) first! We will reply as soon as possible.

It's your turn! Have Fun!

